Title: EDGE-AIT: AI-DRIVEN INDUSTRIAL TRANSPILER GATEWAY

Theme:Hardware(IOT)

Team Name:Nexus Arc

1. The Overview

- **Team Nexus Arc** | Karpagam Academy of Higher Education, Coimbatore
- **Edge-AIT:** A "Digital Brain" for unconnected industrial machines
- **Impact:** Modernizing **14 Million** legacy factory machines across India

- **The Problem**: Factories running "blind" because old machines cannot share data
- **The Innovation:** AI that automatically translates machine signals without any manuals
- **The Security**: 100% Private AI that works without the internet or cloud

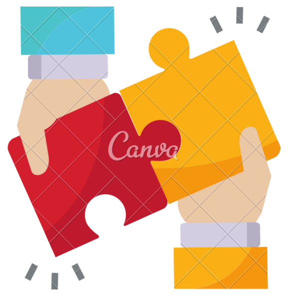
2. The Problem

- **Connectivity Gap:** 70% of Indian industrial machines are legacy systems without data output.
- **Financial Barrier:** Complete hardware replacement costs $150k - $500k, which is unfeasible for MSMEs.
- **Complexity:** Documentation for 1980–2010 era machines is often missing, making manual integration impossible.

- **Target Segment:** Indian MSMEs in Textile, Automotive, and Pharma sectors.
- **Pain Point:** Lack of real-time monitoring leads to 14% energy waste and frequent production downtime.
- **Opportunity:** Modernizing high-value mechanical assets that lack digital intelligence

- **Interpreting Legacy Data:** Edge-AIT functions as a bridge to translate antiquated protocols (Modbus/RS485) into actionable insights.
- **Edge Intelligence**: Uses NVIDIA Jetson Orin Nano for local AI inference, delivering alerts in under 400ms.
- **Data Privacy:** An Air-gapped architecture ensures all industrial data is processed and stored locally within the factory

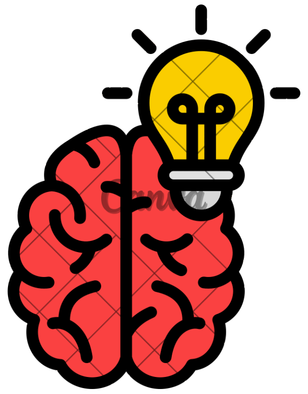
3.The Solution



- **Universal Connectivity:** A plug-and-play gateway that integrates with 90% of PLCs made since 1980.
- **Non-Invasive Setup:** Connects directly to existing RS-485 wiring without requiring factory downtime or machine modification.
- **Industrial Compliance:** Operates within sub-400ms latency, meeting the IEC 61131-3 standard for real-time monitoring.

- **Cost Efficiency:** Hardware implementation at $500 per unit, compared to $150k+ for machine replacement.
- **Operational Savings:** Targeted 50% reduction in downtime and 14% cut in energy waste.
- **Local Reliability:** Operates entirely on-site with zero cloud dependency, eliminating subscription-based data exposure.

- **The Tech Stack:** Powered by NVIDIA Jetson Orin Nano (67 TOPS) running a quantized Phi-3 Mini model.
- **The Intelligence:** Autonomous Register Discovery — AI identifies machine parameters by analyzing real-time signal patterns, bypassing the need for missing manuals.
- **The Security**: Features a Hardware-level Data Diode and AES-256 encryption within a zero-trust architecture.

4. The innovation 

- **Instant Modernization:** Transforms 30-year-old mechanical assets into "Smart Machines" without internal modifications.
- **Autonomous Setup:** Eliminates the need for expensive system integrators or specialized technical manuals.
- **Low-Cost Accessibility:** Provides high-end Industrial IoT (IIoT) capabilities at a fraction of standard market prices.

- **Zero Downtime:** Implementation happens in real-time, preventing the typical $7M/hour industry loss during upgrades.
- **Energy Efficiency:** Real-time power monitoring helps reduce overall factory energy consumption by up to 14%.
- **Predictive Maintenance:** Identifies mechanical fatigue early, extending the lifespan of high-value legacy assets.

Include explainer videos, demos, product screenshots
Include PoCs
Include testimonials of actual customers

5. Market & Opportunity

- **Total Addressable Market (TAM):** The global Industrial AI market is valued at $30 Billion, with a focus on cost-efficient shop-floor digitization.
- **Global Context**: Millions of high-value legacy assets worldwide require low-cost retrofitting to stay competitive in the Industry 4.0 era.

- **Serviceable Market (SAM):** India has 60+ Million MSMEs, of which approximately 12 Million are core manufacturers (Automotive, Textile, and Pharma).
- **The Gap**: Within this sector, 14 Million legacy machines operate without any digital connectivity, forming our primary integration market.

- **Target Market (SOM):** Initial deployment focused on 1.5 Million machines in high-density industrial clusters.
- **Economic Reality:** Replacing a single legacy machine costs $150,000+, making Edge-AIT's retrofitting approach the only viable upgrade path for SMEs.
- **Validated Impact:** Research confirms that the lack of real-time monitoring leads to a 14% loss in operational efficiency for traditional factories.

6. The Technology/Innovation

- **Current Status:** Core software-hardware integration completed and validated in laboratory settings.
- **Technology Readiness Level (TRL):** TRL 4 — Component validation in a laboratory environment, with successful end-to-end signal processing.
- **Development Stage:** Optimizing the quantized Phi-3 model to run within the memory constraints of the Jetson Orin Nano.

- **IP Component:** Proprietary Agentic Discovery Algorithm — A unique pattern-inference logic that maps machine registers without manuals.
- **Innovation:** First-in-class deployment of a 3.8B parameter LLM on edge devices for Autonomous Industrial Diagnostics.
- **Security Architecture:** Features an Air-Gapped framework with local data encryption, ensuring zero external exposure.

- **Technical Roadmap:** * Phase 1 (Current): Lab validation of the "Agentic Discovery" algorithm on Modbus/RS485.
- **Phase 2 (Q3 2026):** Field testing (TRL 5/6) in high-volume textile manufacturing units.
- **Phase 3 (2027):** Commercial release of the plug-and-play AI Gateway for MSMEs.

7. Competitive Landscape

- **Primary Competitors**: Industrial IIoT giants (Siemens, ABB) and established IoT platforms like PTC ThingWorx.
- **Our Advantage:** Zero reliance on Cloud or Internet; works on unsupported legacy protocols where major players require expensive upgrades.
- **Cost Difference:** Traditional SCADA/MES modernization costs $150k+, while Edge-AIT provides a similar insight layer at a fraction of the cost.

- **The Pitch**: We don't ask factories to replace their machines; we give their machines a Local Brain.
- **Edge Advantage:** Unlike competitors who require complex manual tagging/documentation, our Agentic Discovery automates mapping via signal inference.
- **Privacy First:** While others stream data to the cloud, our Air-Gapped architecture ensures critical industrial data never leaves the factory floor.

- **Barriers to Entry:** Proprietary pattern-inference algorithms and optimized Quantized LLM deployment specifically for Jetson hardware.
- **Strategic Partnerships:** Targeting collaborations with Used Machinery Dealers and Industrial Maintenance Providers for rapid distribution.
- **Scale-Up Opportunity:** Eligible to apply for the NVIDIA Inception program to access high-end compute resources and technical mentorship.

8. LEAN CANVAS

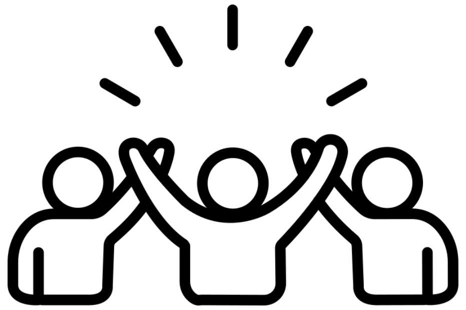
- **Problem:** 70% of legacy machines are digitally "blind"; replacement costs are prohibitive ($150k+).
- **Solution:** Edge-AIT — A plug-and-play AI gateway providing real-time analytics for legacy hardware.
- **Unique Value Proposition (UVP):** Affordable Industry 4.0 for MSMEs via air-gapped, documentation-independent AI diagnostics.
- **Unfair Advantage:** Proprietary Agentic Discovery algorithm for autonomous register mapping and signal inference.

- **Customer Segments:** MSME manufacturing units in Textile, Automotive, and Pharma sectors.
- **Channels:** Direct B2B sales, Industrial
- Maintenance Providers, and used machinery dealers
- **Revenue Streams:** One-time hardware sale (₹55,000) + Annual Maintenance
- **Contract:** (₹1,80,000/year) for AI updates and technical support.

9. Business Model

- **Direct Hardware Sales:** One-time procurement cost of ₹55,000 per unit (Edge-AIT Gateway).
- **AI-as-a-Service (SaaS):** Annual Maintenance Contract (AMC) at ₹1,80,000/year per unit for:
- Continuous AI model optimization and security patches.
- Real-time dashboard access and predictive alert systems.
- Periodic on-site technical calibration.
- **Custom Integration:** One-time project fees for large-scale, factory-wide protocol mapping.

- **Scalable Reach:** Expansion strategy leverages Industrial Maintenance Partners and Used Machinery Dealers to access 200+ units by Year 2 without a massive internal sales team.
- **High Retention:** Once the "Agentic Discovery" maps a machine, the daily operational dependency ensures high AMC renewal rates.
- **Low Entry Barrier:** Unlike expensive $150k+ upgrades, our pricing allows MSMEs to digitize with zero debt.
- **Projected ROI:** Rapid adoption path with a target to hit profitability within 18 months of full-scale rollout

10. The Team 

Kanishk (Project Lead): Systems Engineer – Lead for Edge-AI integration & Hardware optimization.

Haswin SK: Systems Security – Specialist in Air-Gapped architectures.

Kavinya & Mithrasree: Technical Research Leads – Protocol mapping & Documentation.
Krishvanth: Market Strategy – Industrial cluster analysis.

Dhanapal R: Institutional Advisor.

Kanishk: Responsible for the end-to-end technical pipeline, from hardware communication to local AI inference logic.

Haswin: Expert in local data encryption and ensuring industrial-grade network security.

Kavinya & Mithrasree: Spearheading the research on protocol mapping and technical documentation for the "Agentic Discovery" algorithm.

Krishvanth: Analyzing manufacturing trends to align the product with industry demands.

Core Focus: Building high-performance, cost-effective automation tools for the global manufacturing market.

- **Problem-Solving:** Focused on bridging the digital gap in legacy machinery through affordable, non-invasive technology.
- **The Vision:** To provide a practical, "Industrial-Grade" solution that enables smart technology on any factory floor without internet dependency.
- **Team Goal:** To demonstrate that high-performance Edge AI can be implemented securely and cost-effectively for real-world industrial impact.

Thank*you*